OMNISENSE

Unlocking the Voice of the Silent Minority

Transforming fragmented feedback into actionable intelligence for tech markets. Ensuring that every dissatisfied voice is heard and acted upon.

THE CHALLENGE: THE ”ACCURACY PARADOX”

MARKET CONTEXT

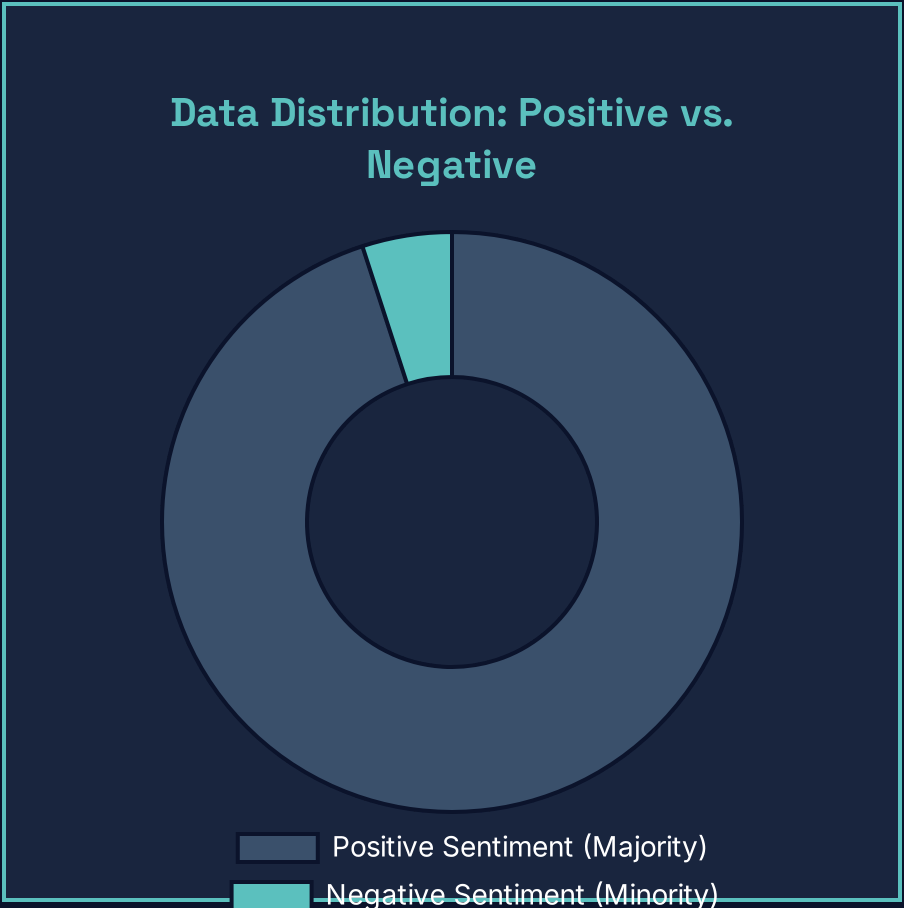
In high-volume tech markets like India and Bangladesh, consumer sentiment is heavily skewed toward positive ratings, creating a "Silent Minority" of dissatisfied users.

THE PROBLEM

Standard machine learning models suffer from the **Accuracy Paradox**—achieving high scores (e.g., 90%+) by simply ignoring the minority of negative reviews.

BUSINESS RISK

Undetected "Negative" sentiment leads to customer churn and brand erosion, as critical product flaws go unnoticed by automated systems.



OMNISENSE VISION: A UNIFIED INTELLIGENCE STREAM

DIVERSE SOURCES

Integrating disparate feedback from Amazon (India) and Mendeley/Pickaboo (Bangladesh) into a single stream.

HARMONIZATION

Implementing dynamic mapping logic to bridge inconsistent naming conventions and data formats.

ACTIONABLE INTEL

Creating a high-recall pipeline that amplifies the "Minority Voice" for strategic decision-making.

> *"Our goal is to transform fragmented customer noise into a unified, actionable intelligence stream that captures every critical pain point."*

DATA STRATEGY: HARMONIZING GLOBAL TECH FEEDBACK

THE CHALLENGE OF HETEROGENEITY

Cross-border data collection introduces significant variation in schema and naming conventions. Amazon India and Mendeley (Bangladesh) use entirely different identifiers for the same semantic content.

Omnisense employs a **Dynamic Mapping Logic** to resolve these inconsistencies, creating a robust, unified record set for downstream modeling.

Unified Record Set
Concatenated & Cleaned Pipeline

SOURCE PLATFORM	ORIGINAL FIELD	UNIFIED TARGET
Amazon (India)	`name`	`unified_text`
Mendeley (Bangladesh)	`Review`	`unified_text`
Pickaboo (Bangladesh)	`comment`	`unified_text`

THE ”ENGINE”: SMOTE-BOOSTED XGBOOST

SMOTE: BALANCING THE SIGNAL

- → **Synthetic Minority Over-sampling:** Generates synthetic examples of negative reviews rather than simple duplication.
- → **Feature Space Interpolation:** Creates new data points along the line segments joining k-nearest neighbors.
- → **Eliminating Bias:** Forces the model to recognize patterns in the minority class that were previously invisible.

XGBOOST: COST-SENSITIVE LEARNING

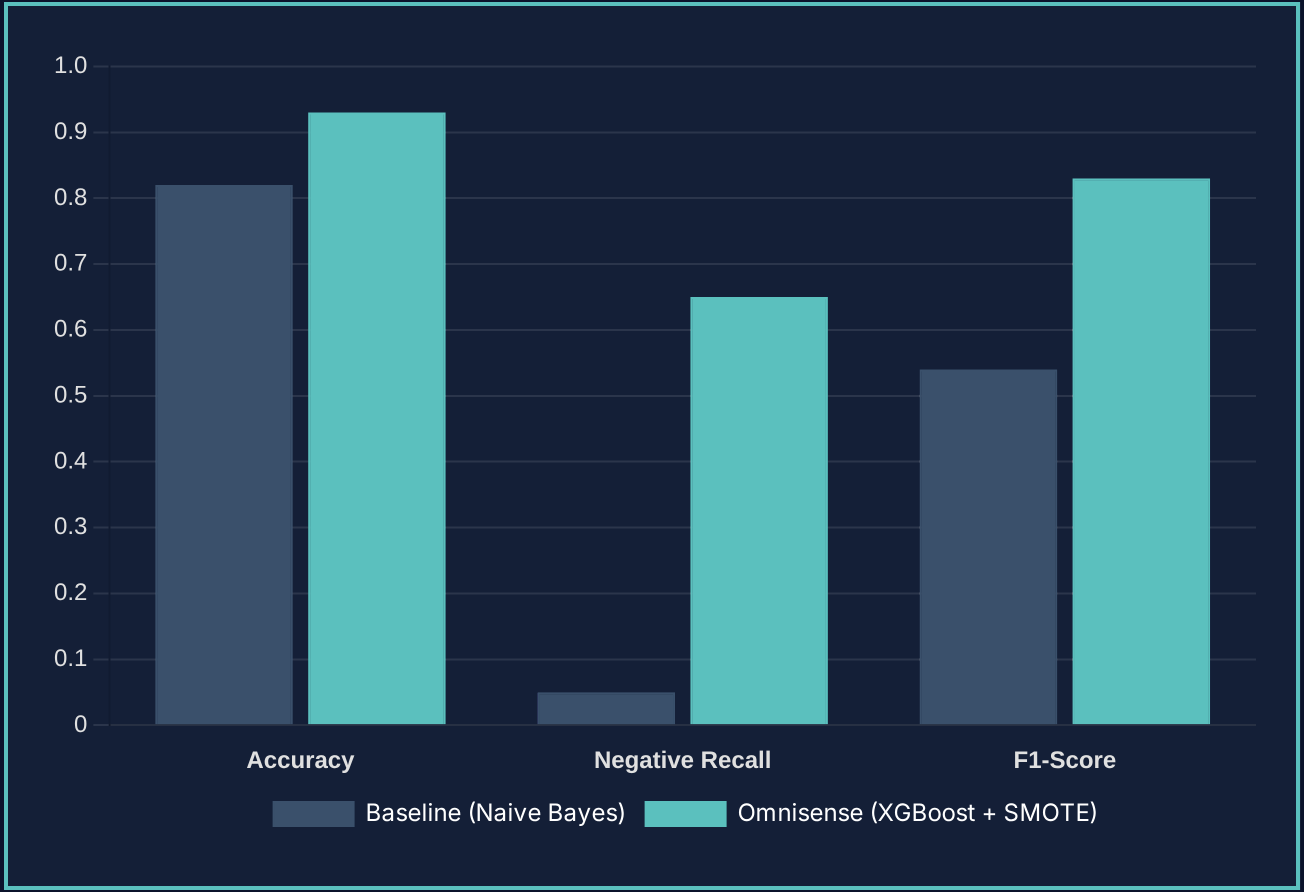
- → **Ensemble Architecture:** Gradient boosted decision trees that iteratively correct previous errors.
- → **Weighted Loss Function:** Penalizes the model more heavily for missing a "Negative" review than a "Positive" one.
- → **High Precision:** Maintains overall accuracy while drastically improving recall for critical feedback.

RESULT: A ROBUST PIPELINE THAT PRIORITIZES CUSTOMER PAIN POINTS OVER SIMPLE MAJORITY ACCURACY.

PERFORMANCE BREAKTHROUGH: OMNISENSE VS. BASELINE

METRIC	BASELINE (NB)	OMNISENSE (XGB)
Overall Accuracy	82%	**93%**
Negative Recall	5%	**65%**
Macro F1-Score	0.54	**0.83**

13X IMPROVEMENT IN NEGATIVE RECALL



INSIDE THE BLACK BOX: EXPLAINABLE AI WITH SHAP

SHAP ANALYSIS

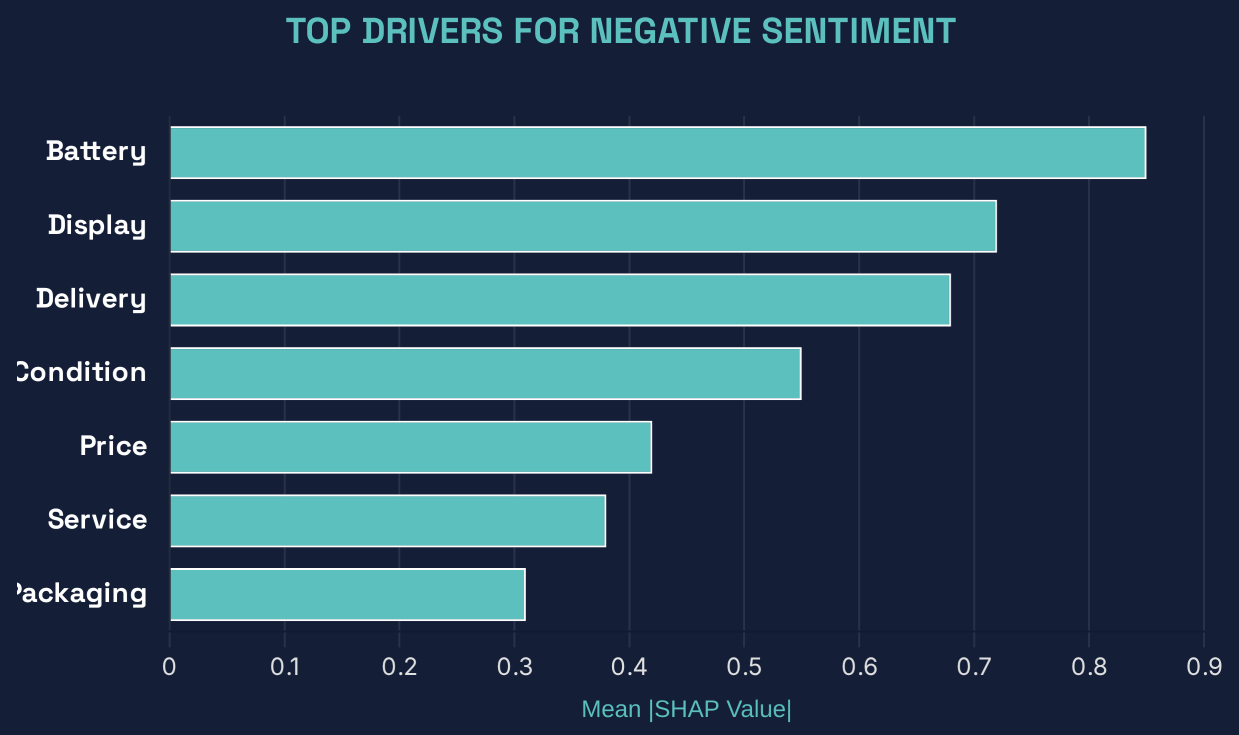
We use SHapley Additive exPlanations to look inside the "Black Box," assigning each feature an importance value for a particular prediction.

THE "WHY" FACTOR

Beyond simple classification, Omnisense identifies the specific linguistic triggers that drive customer dissatisfaction across regions.

TRANSPARENCY

By visualizing global feature importance, we provide stakeholders with evidence-based insights into product and service failures.



STRATEGIC ROI: THE BUSINESS IMPACT MATRIX

DRIVER	ISSUE TYPE	ACTION OWNER	PRIORITY
Battery / Display	Product Quality	R&D / Manufacturing	**HIGH**
Delivery / Condition	Logistical	Supply Chain Ops	**MEDIUM**
Price / Value	Marketing	Strategy / Pricing	LOW

Omnisense transforms raw sentiment into a cross-departmental roadmap. By identifying high-priority technical failures, we enable R&D to focus on the most critical product improvements first.

RECOMMENDATIONS: A ROADMAP FOR CUSTOMER EXCELLENCE

PHASE 01

LOGISTICAL AUDIT

Conduct a deep-dive audit of last-mile delivery partners in Bangladesh to address "Delivery" and "Condition" complaints identified by the model.

PHASE 02

QA FEEDBACK LOOP

Feed "Battery" and "Display" performance data directly back to R&D and Manufacturing to adjust specifications for the next product cycle.

PHASE 03

AUTOMATED ESCALATION

Integrate the Omnisense API into the CRM to automatically flag reviews for immediate response if negative sentiment confidence exceeds 0.8.

OMNISENSE: A COMPETITIVE ADVANTAGE

PROACTIVE INTELLIGENCE

Moving from reactive monitoring to a proactive engine that anticipates customer churn.

GLOBAL SCALABILITY

A flexible pipeline ready to scale across new regions and real-time streaming data.

EMPOWERED VOICE

Ensuring that the "Silent Minority" drives the next generation of product excellence.

EMPOWERING BRANDS TO LISTEN TO EVERY VOICE